

Complex Analysis – 5

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диска

Домашнее задание

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Теоретический минимум. The maximum modulus principle; the Schwarz lemma and its equality case; Schwarz–Pick and the hyperbolic metric of the disc; automorphisms of the disc and of the half-plane; Blaschke factors and finite Blaschke products; Hadamard’s three-circles and three-lines theorems; the Phragmén–Lindelöf principle in a sector.

1. Prove the Schwarz lemma together with its equality case, and deduce Schwarz–Pick by composing with automorphisms; check that the hyperbolic distance $\rho(z, w) = \log \frac{1+|b_w(z)|}{1-|b_w(z)|}$ does not increase under holomorphic self-maps.
2. (S.T. Yau CSMC 2010) Let f be holomorphic on $\mathbb{D}$ with $|f(z)| \leq 1$. Suppose z_0 and $-z_0$ are both zeros of order m of f and $0 < |z_0| \leq \frac{m-1}{m}$. Prove that $|f(0)| < e^{-2}$.
3. (S.T. Yau CSMC 2012, team) Let $V \neq \mathbb{C}$ be a simply connected region, let $a \neq b$ be points of V , and let φ_1, φ_2 be biholomorphic maps of V onto itself with $\varphi_1(a) = \varphi_2(a)$ and $\varphi_1(b) = \varphi_2(b)$. Show that $\varphi_1 \equiv \varphi_2$.
4. (S.T. Yau CSMC 2015, team) Let $U \subset \mathbb{C}$ be a bounded domain with $0 \in U$ and let $f: U \rightarrow U$ be holomorphic with $f(0) = 0$ and $f'(0) = 1$. Prove that $f(z) = z$. (*Подсказка: теорема Картана — посмотрите на итерации f и на коэффициент при первом не исчезающем члене.*)

5. (S.T. Yau CSMC 2020) Let $\alpha \in \mathbb{R}$ and $a_1, \dots, a_n \in \mathbb{D}$, and consider

$$f(z) = e^{2\pi i \alpha} z \frac{z - a_1}{1 - \bar{a}_1 z} \cdots \frac{z - a_n}{1 - \bar{a}_n z}.$$

Prove that f maps the unit circle onto itself and preserves the measure $d\theta$ on it.

6. (S.T. Yau CSMC 2015) Describe all functions holomorphic on $\mathbb{D}$, continuous on $\bar{\mathbb{D}}$ and mapping $\partial\mathbb{D}$ into $\partial\mathbb{D}$.
7. Prove Hadamard's three-circles theorem: for f holomorphic in the annulus $r_1 < |z| < r_2$ the function $\log M(r)$ is a convex function of $\log r$. Deduce the three-lines lemma by an exponential change of variable.
8. (S.T. Yau CSMC 2025) Let $\Omega = \{x + iy : x > 0, y > 0\}$ and let f be bounded and continuous on $\bar{\Omega}$, holomorphic on Ω , with $\max\{|f(x)|, |f(ix)|\} \leq 2$ for $x \leq 1$ and ≤ 1 for $x > 1$. Prove that $|f(x + iy)| \leq 2^{\frac{2}{\pi} \arctan \frac{1}{2xy}}$ on Ω .
9. Prove the Phragmén–Lindelöf principle for a sector of opening π/β : if f is holomorphic in the sector, continuous up to the boundary, $|f| \leq M$ on the boundary and $|f(z)| \leq Ce^{c|z|^\rho}$ with $\rho < \beta$, then $|f| \leq M$ inside. Show by an example that the growth restriction cannot be dropped.
10. (границная лемма Шварца) Let $f: \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$, and suppose f extends holomorphically past the boundary point 1 with $f(1) = 1$. Prove that $f'(1) \geq 1$, and that $f'(1) = 1$ only for $f = \text{id}$.